Skill Issue – User Manual

Peer-to-peer skill exchange: teach, learn, earn points.

1. Requirements

g++ 9+ (C++17) and make; Node.js 18+ only for the web UI.

2. Build

```
make         # builds ./a.out (console) and ./server (web API)
make test    # optional: 59 automated tests
```

3. Run

```
./a.out                                      # console version
./server                                     # web API, http://localhost:8080
cd frontend && npm install && npm run dev    # web UI,  http://localhost:5173
```

Start from the project folder so that `data/` (the CSV files) is found.

4. Using the app

1. **Register** with ID, name, password and skills separated by `;` (e.g. `C++;Python;Guitar`).
2. **Log in** (console admin: `admin` / `admin123`).
3. **Learn:** pick a skill, choose a mentor from the ranked list, enter a preferred time, press *Book*.
4. **Teach:** open *Mentor Inbox*; *Accept* with a proposed time or *Reject*.
5. **Confirm:** in *My Bookings* accept the mentor's offer; a session is scheduled.
6. **After the session:** mentor reports Completed/Missed/Cancelled; learner rates the mentor (1–5); mentor grades the quiz.
7. **Leaderboard** ranks mentors by teaching points; **Profile** shows points and lets you edit skills.

5. Points

Rating ×5 gives the mentor teaching points (console); quiz marks give the learner points; a confirmed no-show costs 5 points; new users start with 30.

6. Troubleshooting

Invalid ID or password	IDs and passwords are case-sensitive.
Web UI cannot reach API	Start `./server` first; port 8080 must be free.
Data looks empty	Run from the folder that contains `data/`.
Reset	Delete the four CSV files in `data/` (keep a backup).

Prototype notice: passwords are stored in plain text – do not use real passwords.